

Core Finance

Week 5

MGMP 543

Prof. Benedict Guttman-Kenney
Spring 2025



RICE | BUSINESS
Jones Graduate School of Business

Hello! Today's Topics

- Week 4 Recap
- Course Admin: Survey, Case, & Midterm
- Capital Asset Pricing Model (CAPM)

Week 5: Last Week of Introductions 🥲

Week 5: Last Week of Introductions

Your Name

Options from previous weeks

- Something interesting/unusual about a place you grew up / have lived
- Best thing done in last week/month/2025
- Favorite food/restaurant
- Unpopular opinion (Popular that is overrated / Popular is underrated)

New options

- Fact about you
- Dream dinner guest
- Where would you like to visit / what thing would you like to do
- Wonderbox
- Random Question

Week 4 Recap

Week 4: What Did We Learn?

Week 4: What Did We Learn?

- Returns on Single-Stocks
- How to Measure Risk
- Returns on Portfolios of Securities
- Risk of Portfolio of Securities
- Efficient Frontiers & Benefits of Diversification
- Limits of Diversification

Quiz 4: Assume that you purchased one share XYZ on the last day of 2023 and sold that share on the last day of 2028. Use this series of stock prices and dividends (you can assume dividends are paid on the date) to answer the quiz questions.

Date (Year-end)	Stock Price (XYZ)	Dividend per share
2023	\$123.50	--
2024	\$154.63	--
2025	\$165.32	\$4.00
2026	\$162.30	\$1.23
2027	\$178.63	--
2028	\$185.50	\$3.23

Q2. What is the total return on your investment? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

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Q2. What is the total return on your investment? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

Total Return is the sum of dividends and capital gains, as a percent of purchase price:

$$\text{\$4.00} + \text{\$1.23} + \text{\$3.23} + (\text{\$185.50} - \text{\$123.50}) = \text{\$70.46}$$

Total Returns as a percent of purchase price:

$$\text{\$70.46}/\text{\$123.50} = 57.05\%$$

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Q3. What is the average arithmetic annual return? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

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Q3. What is the average arithmetic annual return? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

$$\frac{154.63 - 123.50}{123.50} + \frac{4.00 + (165.32 - 154.63)}{154.63} + \frac{1.23 + (162.30 - 165.32)}{165.32} + \frac{(178.63 - 162.30)}{162.30} + \frac{3.23 + (185.50 - 178.63)}{178.63}$$

$$= 9.87\%$$

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Q4. What is the sample variance of your annual return? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

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Q4. What is the sample variance of your annual return? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

Step 1. Calculate the annual return for each year.

For example, return in 2025 is $[4.00 + (165.32 - 154.63)] / 154.63 = 9.5\%$

Step 2. Calculate sample variance for each year

For example, sample variance in 2025 is $(0.09500097 - 0.09867917)^2 = 0.00001$, where 0.09867917 is the average arithmetic annual return calculated in question 3.

Step 3. Sum up sample variances and divide by N-1. N=5 (five years of returns) so divide by 4.

0.93%

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Q5. What is the portion of your return earned through capital gains? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

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Q5. What is the portion of your return earned through capital gains? Give your answer as a percentage, rounded to two decimal places. For example, if your answer is 1.234% write 1.23

$\$185.50 - \$123.50 = \$62.00$ in capital gains

$\$8.46$ in dividends

$\$62.00 / (\$62.00 + 8.46) = 88.00\%$

Survey, Case, & Midterm

Mid-Course Feedback Survey

Please complete this!

Optional, anonymous and not graded assignment.

- Feedback on teaching
- Custom Week 11 topics/questions to cover (e.g., consumer finance, bank runs, great recession)
- Evaluate interest in a new 6 week elective course on FinTech

Please complete by Monday 24 February

Beta Management Case

Complete in your usual MBA groups (on canvas)

Case write-up is due before start of class next week.

AI tools must not be used for this or any other graded assignments.

Submit your case write-up on canvas once per group. Write all names of group members at the top.

Be prepared to present your findings in-class.

All group members should be prepared to answer questions on the case in class.

Midterm

Reese (Teaching Assistant) will be going over practice questions in his zoom office hour Sunday 23rd at 4pm CT.

Midterm Details

Duration: You have 2 hours to complete this.

Content: Covers all topics across first 6 weeks.

Open Book: You may use any textbook, your own notes, Excel, and any materials posted to Canvas.

I will provide “Cheat Sheet” with main formulas.

You may not receive assistance from any person or website. Failure to abide by this will result in a zero. AI tools must not be used for this or any other graded assignments.

When Available? Exam is released after your section’s week 6 class finishes.

Deadline? Exam must be completed by

- 6pm CT Monday 31 March for Monday Section / by 6pm CT Wednesday 2 April for Wednesday Section

You can take the exam at any time in that window so long as you complete it by the above deadline.

No late submissions are accepted. You will get a grade of 0.

The last question of the quiz allows you to upload a file showing your workings. This must be done within the quiz time limit to be considered – no late submissions via email (or other method) will be accepted.

Cost of Equity

Where r you coming from?

Where do **discount rates** come from?!

$$PV = \sum_{t=0}^{\infty} \frac{C_t}{(1 + r)^t}$$

In the real world, the discount rate isn't given to you!

How to determine the correct discount rate?

1. Use the capitalization rate (week 3 class)

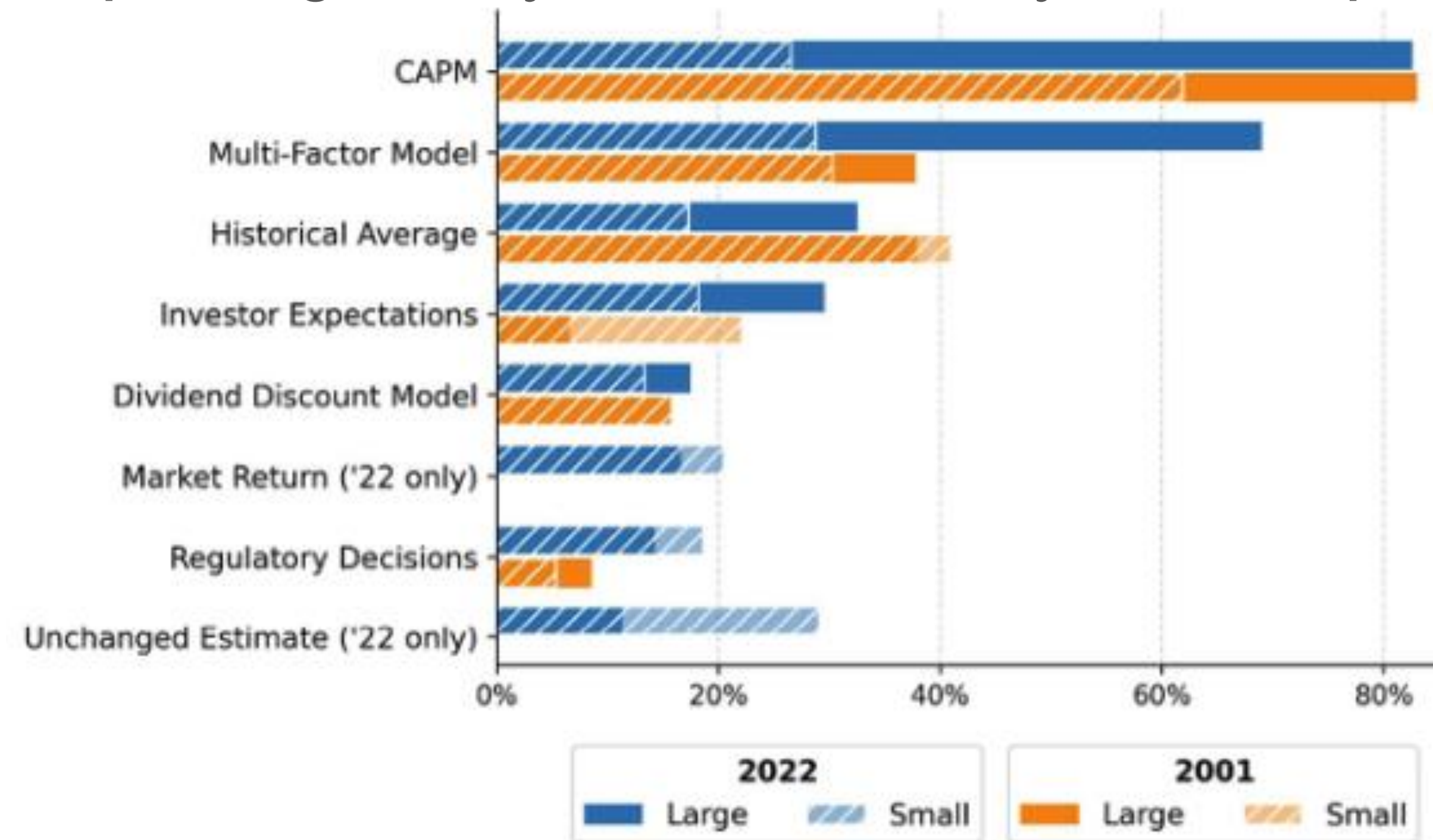
$$r = \frac{D_1}{P} + g = \text{dividend yield} + \text{growth rate}$$

2. Use historical data (week 4, last class)

3. Use asset pricing models (this week's class)

How Do Companies Actually Estimate the Cost of Equity?

Percent of CFOs reporting always/almost always use a particular approach



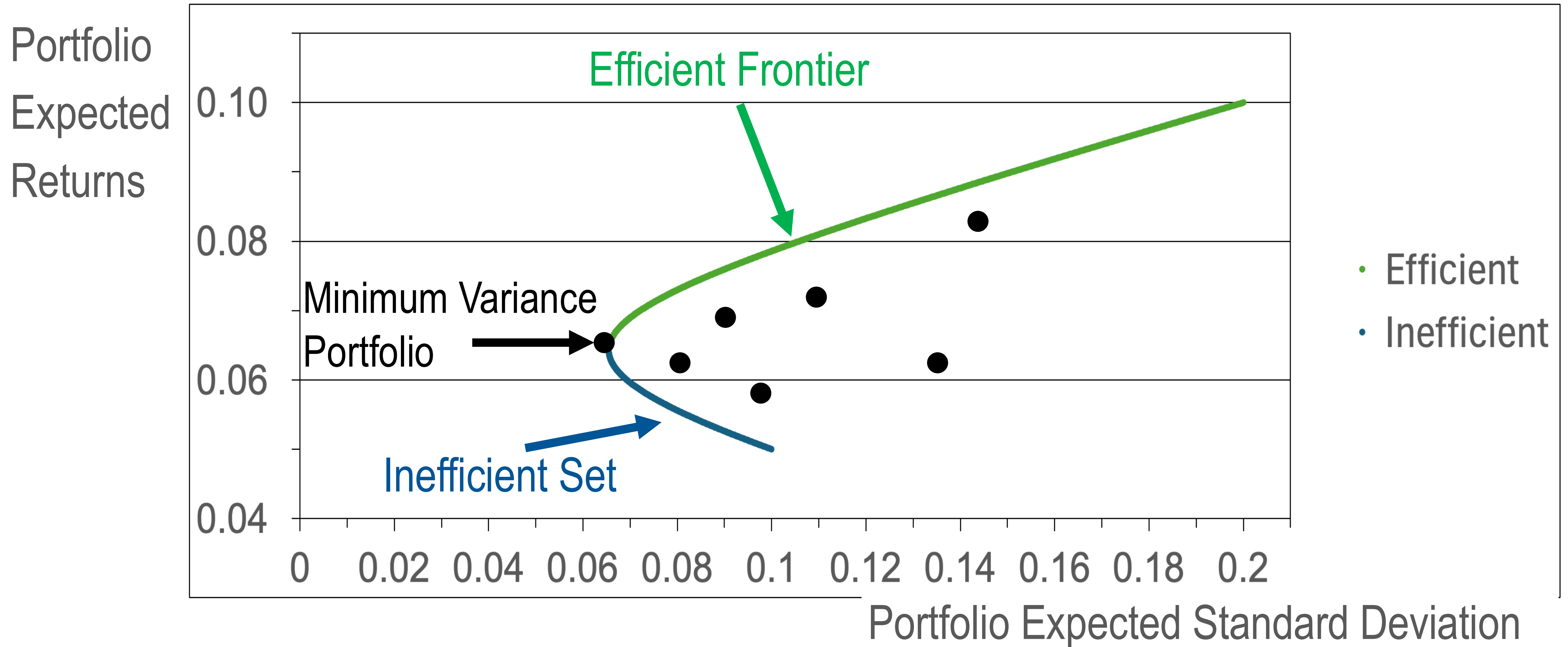
John Graham (2022).

Large firms are \$1bn+ revenue.

CAPM used more by larger firms, leveraged firms, public companies, CEO has MBA
Need to know cost of equity for capital budgeting (are potential projects $NPV > 0$?)

Portfolios with a Risk-Free Asset

Our Efficient Frontier with Risky Assets



Efficient Frontier With Risk-Free Asset

“Risky” Asset (e.g., Tesla stock) “i” with expected return r_i and σ_i s.d.

“Risk-Free” Asset (e.g., US government debt) “rf” with expected return r_{rf} and 0 s.d.

Expected Portfolio Return:

$$E[r_p] = w_{rf}r_{rf} + w_iE[r_i]$$

where $w_{rf} + w_i = 1$

Efficient Frontier With Risk-Free Asset

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Expected Portfolio Return:

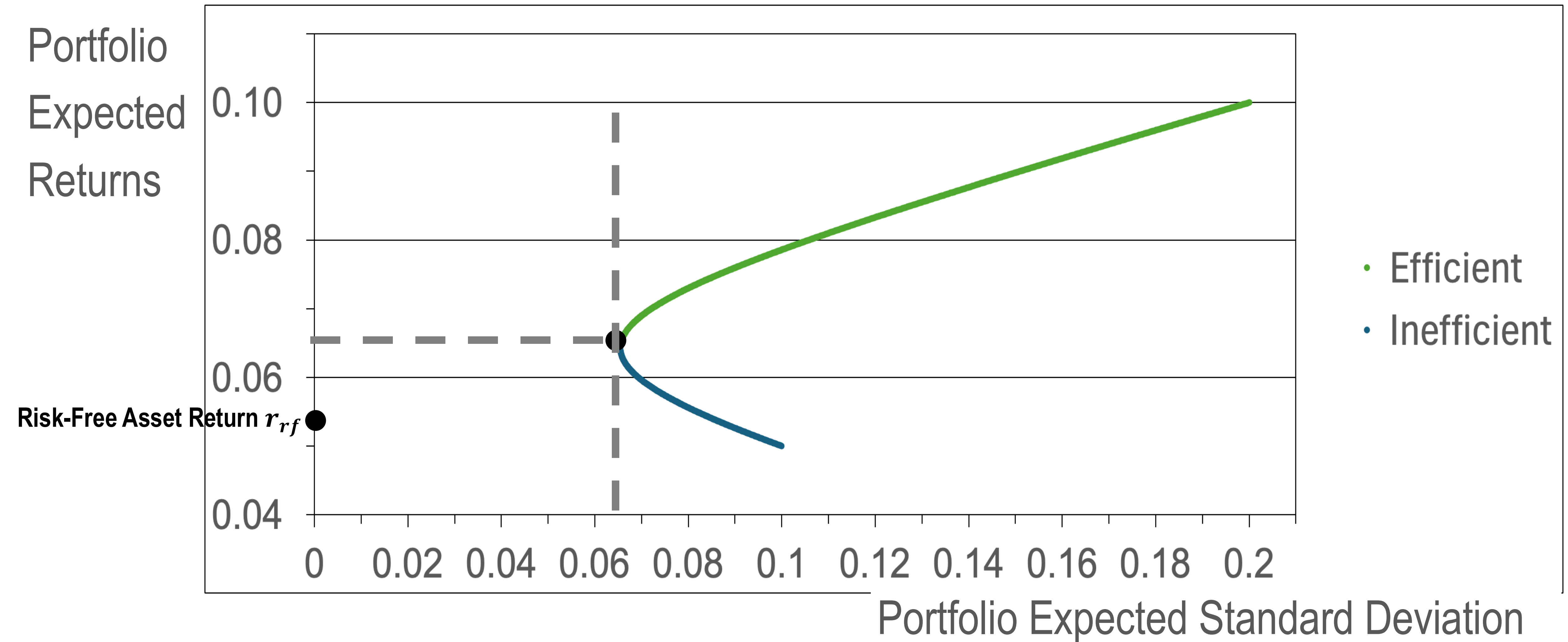
$$E[r_p] = w_{rf}r_{rf} + w_iE[r_i]$$

$$\text{where } w_{rf} + w_i = 1$$

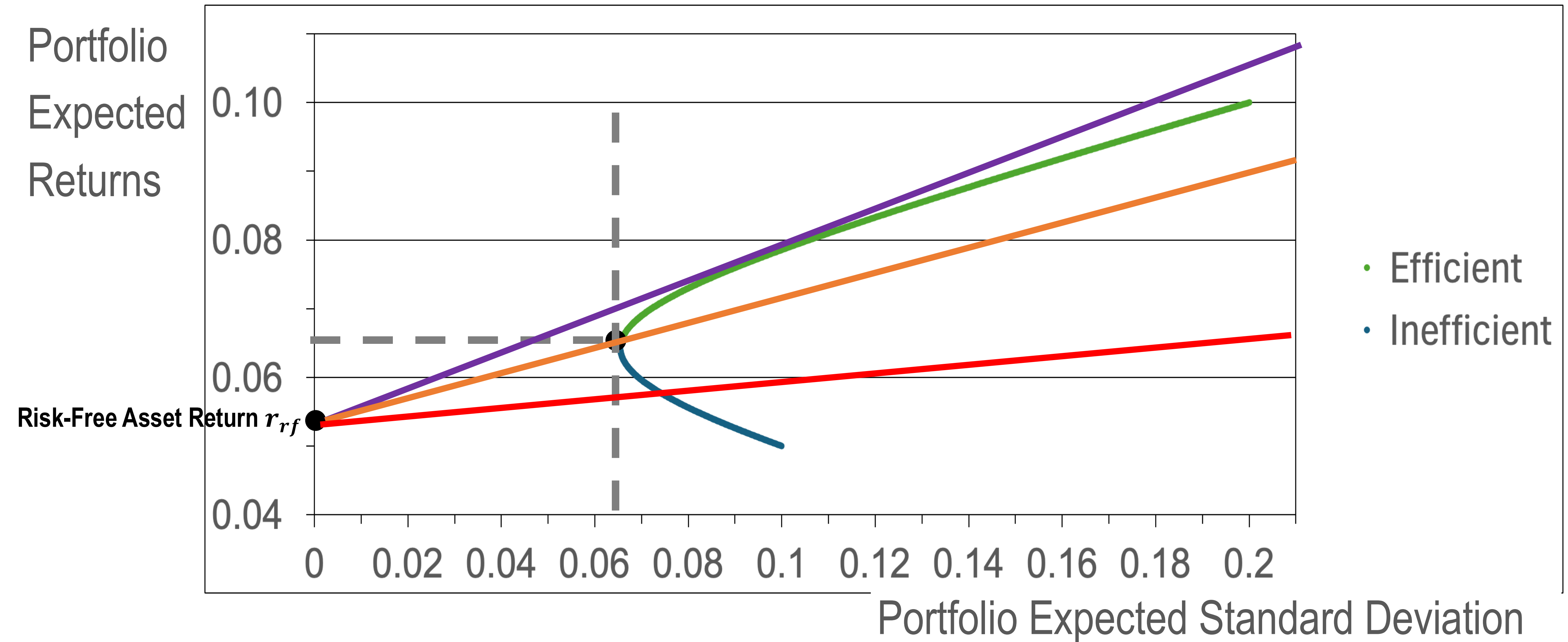
Expected Portfolio Risk:

$$\sigma_p = \sqrt{w_{rf}^2\sigma_{rf}^2 + w_i^2\sigma_i^2 + 2w_{rf}w_i\rho_{rf,i}\sigma_{rf}\sigma_i} = w_i\sigma_i$$

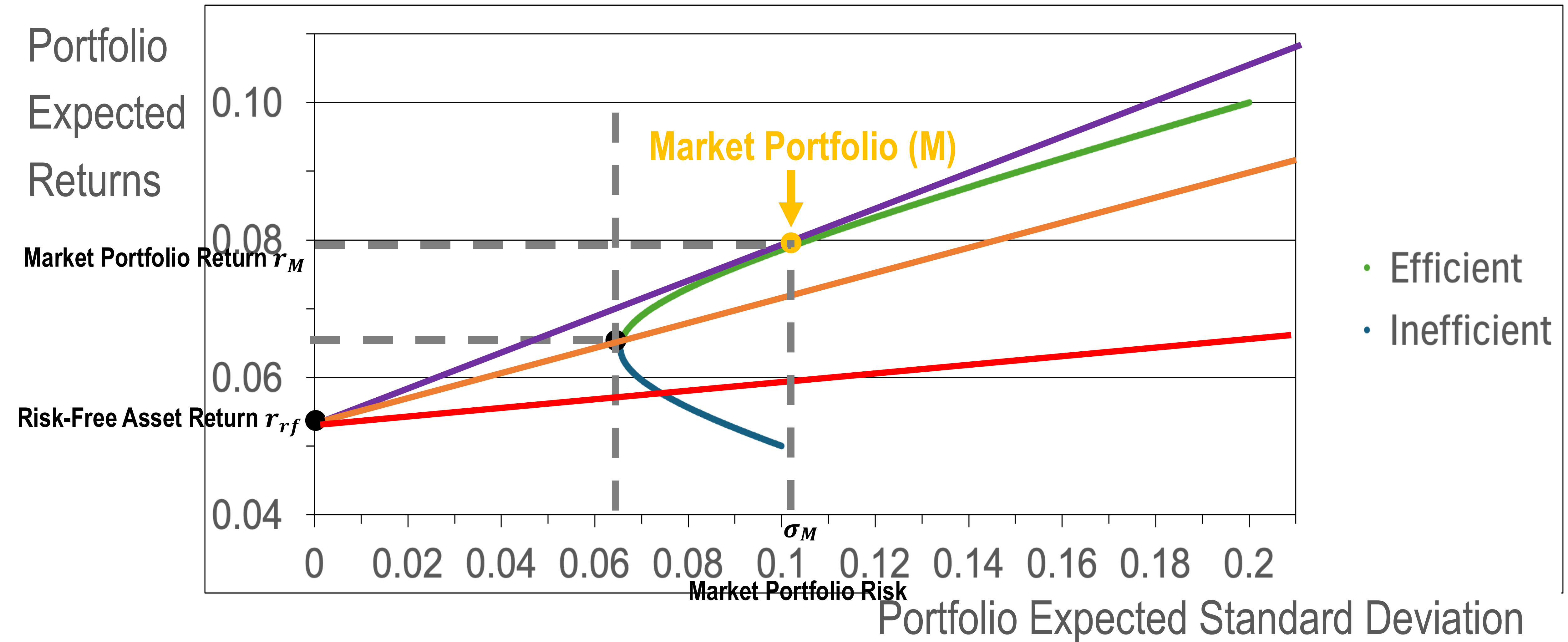
Expected Return & Risk of Portfolio With Risky Asset and Risk-Free Asset



Expected Return & Risk of Portfolio With Risky Asset and Risk-Free Asset



Market Portfolio is on a Tangency Line to Efficient Frontier Running from the Risk-Free Asset



Any combination of the Tangency portfolio and risk-free asset maximizes the Sharpe Ratio

Sharpe Ratio is the slope of line combining a risky asset and the risk-free asset.

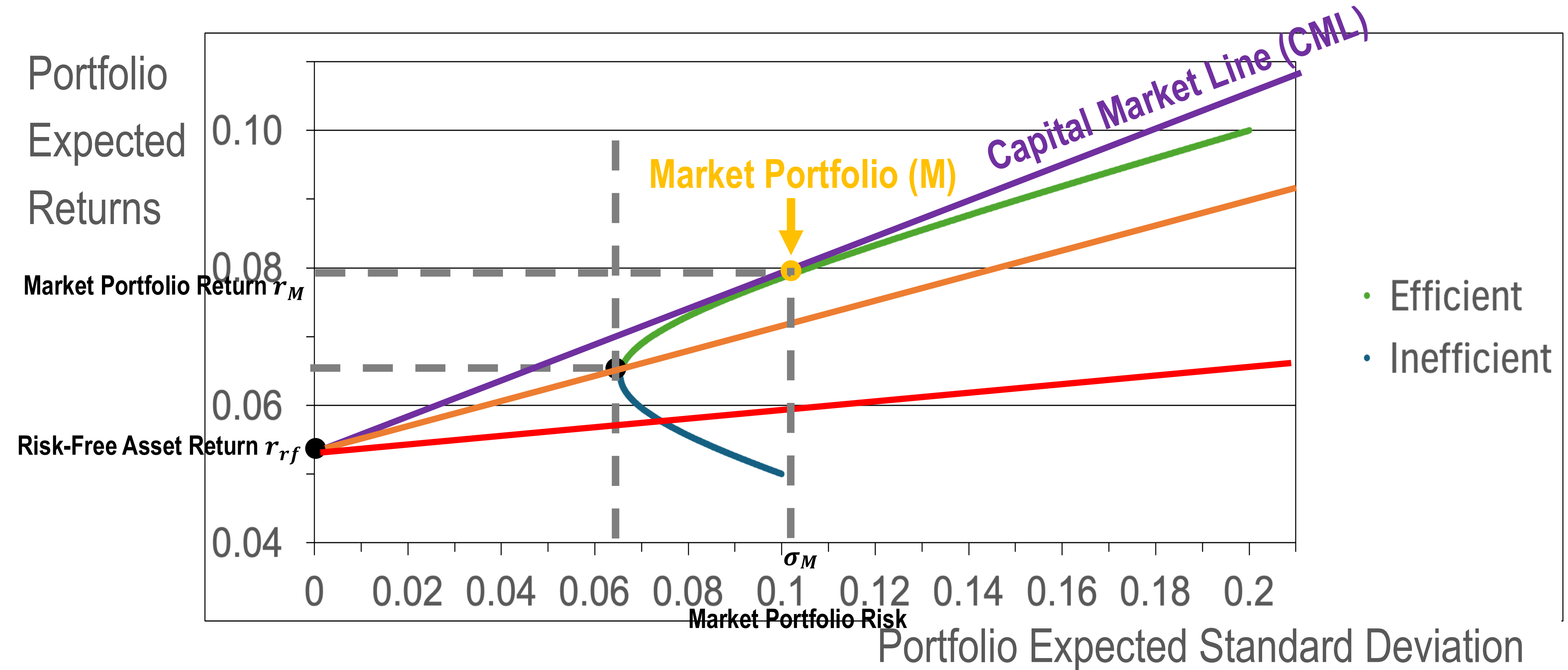
Measures how much return on a stock in excess of risk-free rate, per unit of risk

Investors always prefer a higher Sharpe Ratio

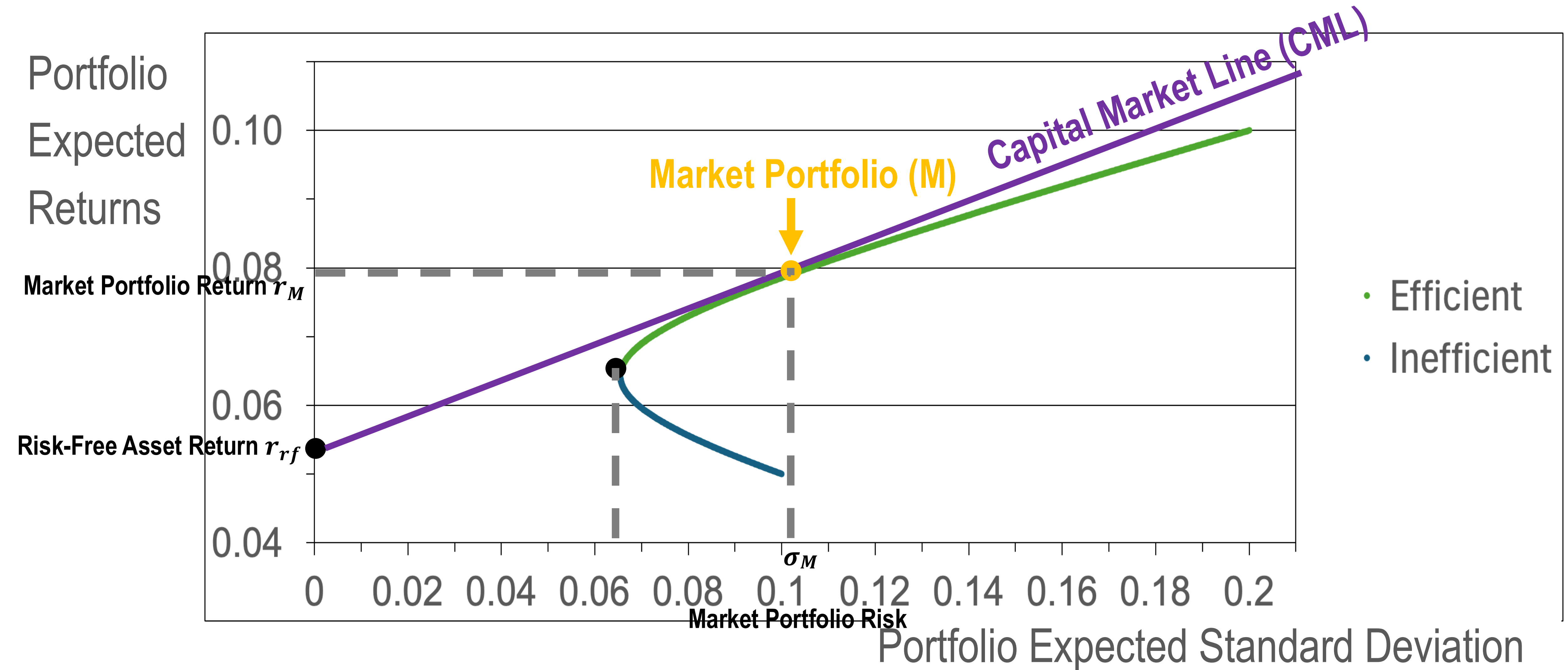
-> Higher risk-adjusted return

$$\text{Sharpe Ratio} = \text{Slope} = \frac{E[r_i] - r_{rf}}{\sigma_i}$$

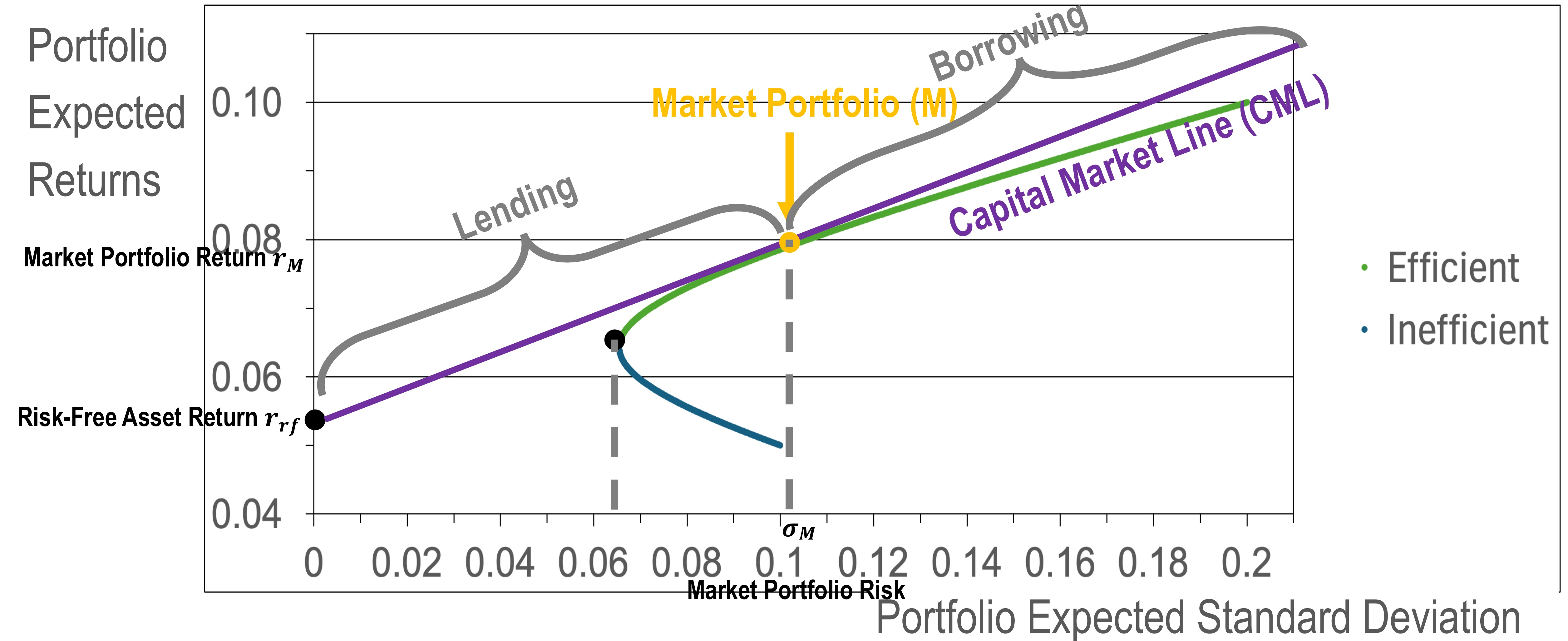
Capital Markets Line (CML) is the name for the Tangency Line to Efficient Frontier Running from the Risk-Free Asset



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Lending and Borrowing



Borrowing and Lending

We need $w_{rf} + w_i = 1$

So, if $w_m = 1$ it means that $w_{rf} = 0$ and $w_i = 1$

And if $w_m = 0$ it means that $w_{rf} = 1$ and $w_i = 0$

Borrowing and Lending

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So, if $w_m = 1$ it means that $w_{rf} = 0$ and $w_i = 1$

And if $w_m = 0$ it means that $w_{rf} = 1$ and $w_i = 0$

But it is also possible to have:

$w_m > 1$ then “borrowing” at the risk-free rate because paying r_{rf} . Means that $w_{rf} < 0$ and $w_i > 1$
(mutt-jargon bustin’: “Buying on margin” / “using leverage”. Borrowing to invest in stocks.)

$w_m < 1$ then “lending” at the risk-free rate because receiving r_{rf} . Means that $w_{rf} > 1$ and $w_i < 0$

‘Separation Theorem’: Borrowing/Lending decision is separate from the choice of portfolio of risky assets.

Slope of Capital Market Line (CML) is the Market Price for Risk for Efficient Portfolios

Reminder: Slope of line combining a risky asset and the risk-free asset is the Sharpe ratio:

$$\text{Sharpe Ratio} = \text{Slope} = \frac{E[r_M] - r_{rf}}{\sigma_M}$$

CML:

$$E[r_p] = r_{rf} + \sigma_p \left(\frac{E[r_M] - r_{rf}}{\sigma_M} \right)$$

CML shows the trade-off between risk and return for efficient portfolios. Where 'M' is the market portfolio, 'P' is the portfolio chosen. CML must slope upwards because higher returns carry more risk.

Measuring Risk of Assets

In market equilibrium, expected return for any asset is equal to risk-free rate + risk premium.

Risk-free rate can be approximated by the yield on government bonds (Treasury Bills)

Risk premium of market portfolio is not observed. Proxied by broad market index e.g., S&P 500 Index.

Capital Market Line (CML)

- Can achieve the same standard deviation as any portfolio on the efficient frontier, but obtain a higher expected return by combining the market portfolio and risk-free asset.
- Where on the line an investor chooses depends on their risk preferences.
- If homogenous (all the same) expectations, all (rational) investors choose the same risky portfolio: the market portfolio.
- Market Portfolio: Each risky asset has weight proportional to its market value.
- All investors will choose a portfolio along the efficient frontier.
- All investors will invest in portfolio on CML because they dominate any portfolio on the efficient frontier.
- All investors will select a combination of M and RF on CML to satisfy their risk tolerances.
- Market portfolio is fully diversified and eliminates all diversifiable risk. Risk-free asset eliminates diversifiable and non-diversifiable risk.

Beta

Measuring Risk of Individual Securities

Idea is if you add a new asset to the market portfolio, how does it affect the portfolio's risk (standard deviation)?

Will depend on covariance between new asset and the market.

β (“Beta”) Measures Relative Riskiness

Beta of security i is:

$$\beta_i = \frac{COV(r_i, r_M)}{\sigma_M^2}$$

Where r_i is the return on security i .

r_M is the return, and σ_M^2 is the variance of the market overall.

β (“Beta”) Measures Relative Riskiness

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$$\beta_i = \frac{COV(r_i, r_M)}{\sigma_M^2}$$

Where r_i is the return on security i .

r_M is the return, and σ_M^2 is the variance of the market overall.

Beta is the individual security's sensitivity to the overall market (covariance), normalized by the overall market variance.

If $\beta_i = 1$, systematic risk is the same size as the market risk. So beta of market portfolio is 1.

If $\beta_i = 0.5$, then if market goes up by 10%, security i expected to go up by 5%.

If $\beta_i = 2$, then if market goes up by 10%, security i expected to go up by 20%.

We can estimate β (“Beta”) in a regression

$$r_{i,t} = \alpha + \beta(r_{M,t} - r_{rf,t}) + \varepsilon_t$$

You can also find this for each stock on Yahoo! Finance / other sources.

Beta is commonly calculated based on monthly returns using 5 years of monthly data. But can calculate over different frequencies (daily/weekly/monthly) and different horizons (1/3/5/10 years) depending on how used.

For example, traders & portfolio managers may be interested in shorter-term returns.

For example, pension funds may be interested in longer-term returns.

Beta of Portfolio of Assets: β_p

Beta of portfolio (β_p) is the weighted sum of betas of stocks in a portfolio

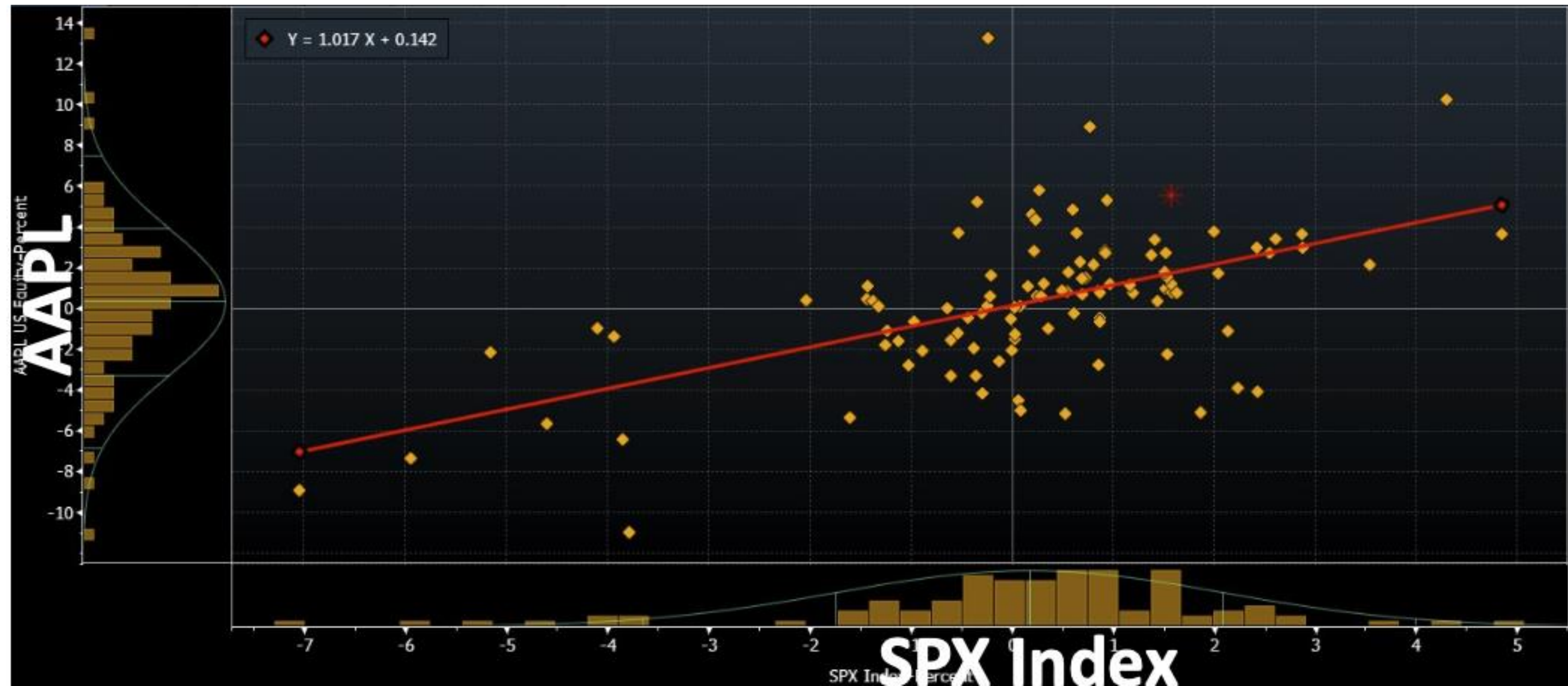
Step 1. Calculate beta for each stock

Step 2. Weight beta for each stock by each stock's share of portfolio

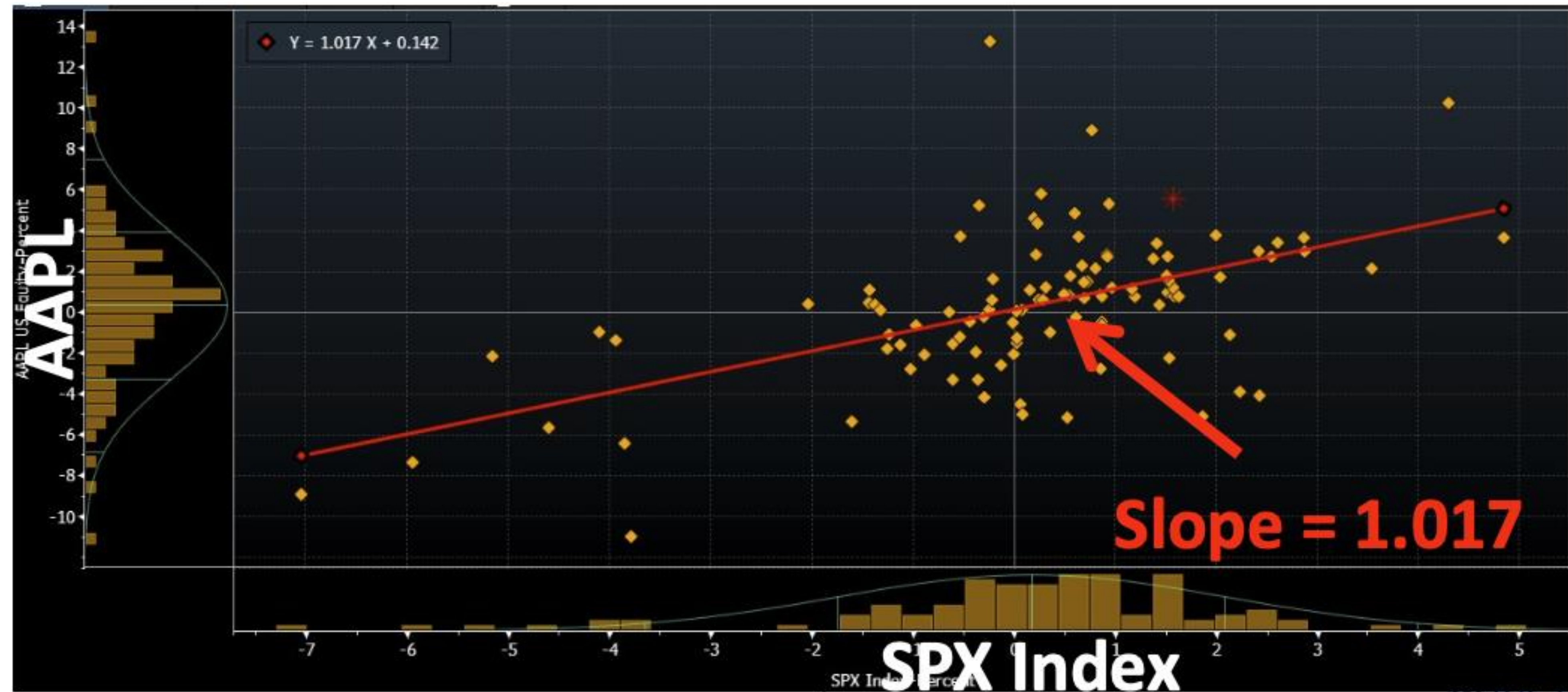
$$\beta_p = \sum_{j=1}^N \beta_j w_j$$

Betas in the Wild

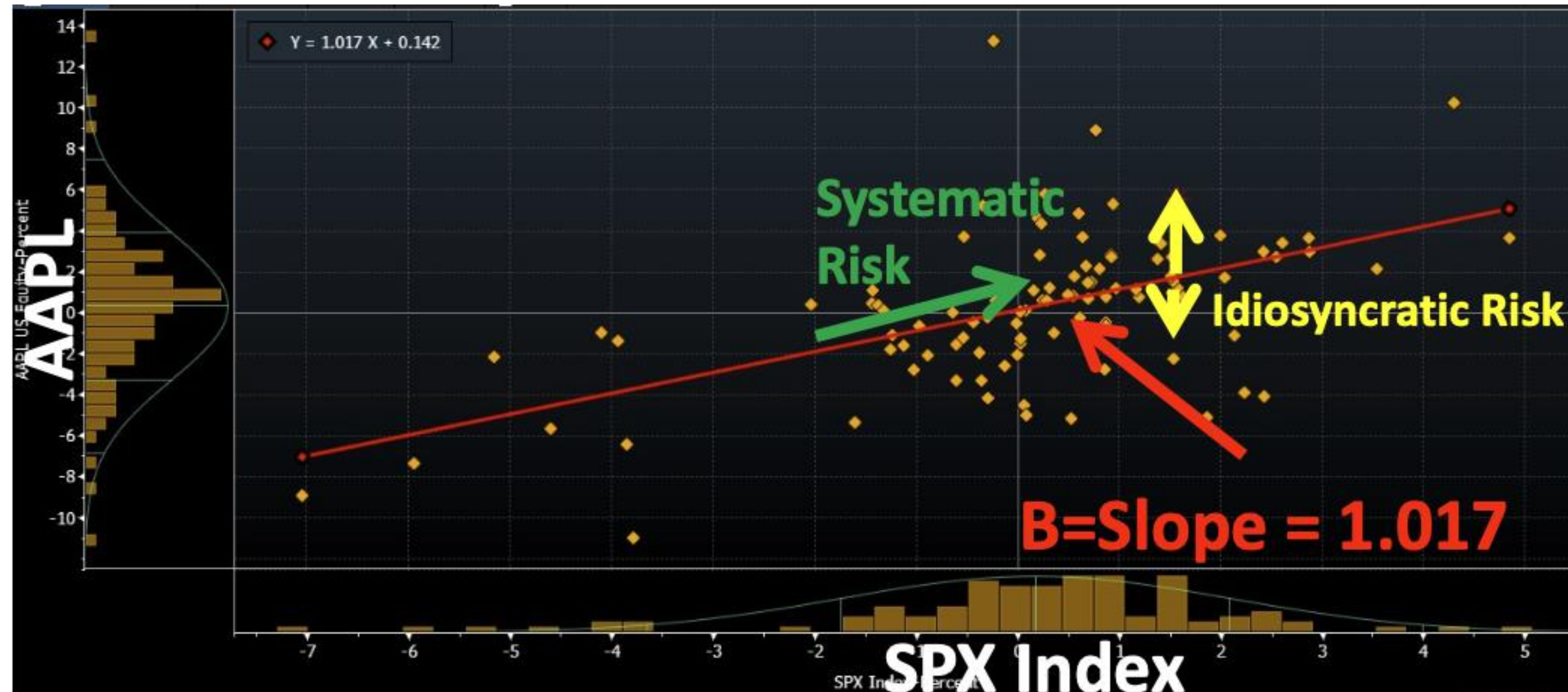
Real-World Beta Example: Comparing Apple (AAPL) to S&P 500 (SPX)



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Real-World Betas (<https://finance.yahoo.com/>)

Company	Beta

Security Market Line (SML)

Security Market Line (SML) / Capital Asset Pricing Model (CAPM)

Security Market Line (SML):

$$E[r_i] = r_{rf} + \beta_i(r_M - r_{rf})$$

Security i's expected return is the risk free rate + risk premium.

Market risk premium is: $E[r_M] - r_{rf}$.

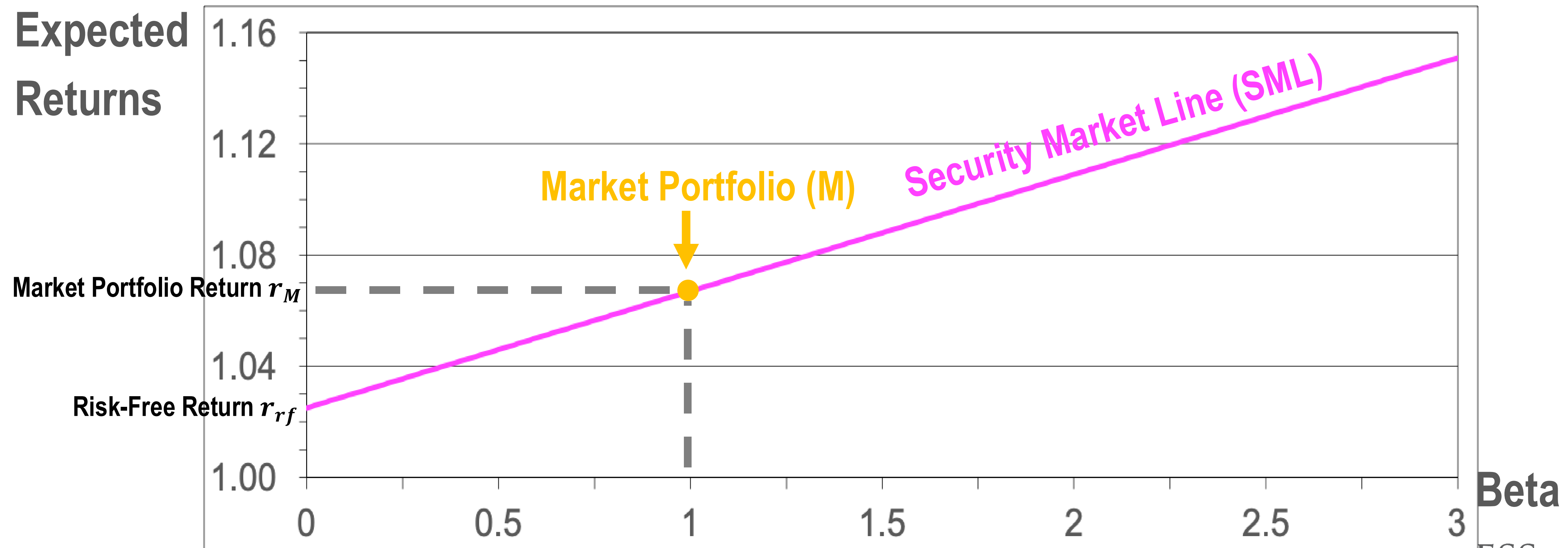
β_i captures the quantity of risk.

Security risk premium for security i is: $\beta_i(E[r_M] - r_{rf})$

Under CAPM, the required rate of return **for any security** is a linear function of the security's risk.

Security Market Line (SML)

$$E[r_i] = r_{rf} + \beta_i(r_M - r_{rf})$$



Is the Capital Market Line (CML) the same as the Security Market Line (SML) the same? NO! (smh)

Capital Markets Line (CML)

- CML explains that the market portfolio is the best portfolio of risky assets
- CML holds for fully diversified portfolio, without unsystematic risk
- CML does not hold for single securities, which have systematic risk

Security Markets Line (SML)

- Given the market portfolio is the best portfolio of risky assets (CML), the SML uses that to determine the required return for any risky asset.
- SML measures systematic risk (beta)
- SML holds for all portfolios and individual securities

Testing CAPM

CAPM predicts that:

- (i) Security prices lie on the SML (linear in beta)
- (ii) Beta is the only reason why expected returns differ across risky assets

Empirical evidence?

Estimated SML appears to be linear but intercept term higher than risk free rate, and SML's slope less steep than theory.

Overall, a good logical way to view risk-return trade-off.

(Jensen's) Alpha Measures Excess Returns

$$r_{i,t} - r_{rf,t} = \alpha_i + \beta_i(r_{M,t} - r_{rf,t}) + \varepsilon_{i,t}$$

Active investors try to find assets that are mis-priced relative to CAPM (or factor models).

“Seeking alpha” <- investment strategy that exceeds a benchmark. Often described as an investment strategy’s ability to be the market (or its “edge”).

Alpha is the excess return or the abnormal rate of return related to a benchmark (after adjusting for risk)

CAPM used not just for cost of equity capital, but to evaluate investment performance.

If CAPM Assumptions do not hold, stocks won't be on SML

1. All investors are price takers
2. All investors have the same horizon
3. All investors have the same information and interpret it in the same way
4. Capital markets are “perfect”. No transaction costs, no taxes, no restrictions on short selling (strategy to profit from decline in security's price. Borrow shares, selling them and aim to buy them back for less money)
5. All investors are risk-averse
6. The market portfolio exists
7. There exists a risk-free rate assets such that investors can borrow or lend unlimited amounts at this risk-free rate.

Factor Models

Factor Models

1 Factor: Market Excess Return

$$r_i - r_{rf} = \beta(r_M - r_{rf})$$

Arbitrage Pricing Theory (APT): Macro factors + noise explain.

$$r_i - r_{rf} = b_1(r_{Factor\ 1} - r_{rf}) + b_2(r_{Factor\ 2} - r_{rf})$$

Fama-French Three Factor Model (1992): Market Excess Return, Size (small companies outperform large, **Small Minus Big**) ~ 3.6% p.a., Book-to-Market (High outperform low, **High Minus Low**) ~5.2% p.a.

Explains ~90% of diversified portfolio returns.

$$r_i - r_{rf} = \beta_i(r_M - r_{rf}) + b_{SMB,i}(r_S - r_B) + b_{HML,i}(r_H - r_L)$$

Fama-French Five Factor Model (2015): Added Operating Profitability (High/robust vs. low/weak, **RMW**), Investment (conservative vs. aggressive, **CMA**)

Momentum: Stocks that perform well in the past tend to overperform those that have performed poorly in the past. Over 1 month (or short-term horizons) opposite may be true as prices revert to means. Short-lived so may not affect cost of equity estimates.

You can download historical factors here: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Implications of Asset Pricing Models for Investments

If CAPM is correct...

You *cannot* beat the market, on average.
Instead, passively invest in diversified index fund,
minimize your fees.

If CAPM is incorrect...

You *can* beat the market, on average.
Make active investments that exploit predictable mis-pricing.

Irrespective of CAPM being correct or incorrect...

Individual investment choices will differ with risk tolerances.

We will discuss more in Week 7 (post-midterm)!

Examples

Q1. Calculate expected return and standard deviation

Year	Tangent Portfolio	Risk-Free Asset
E(r)	14%	6%
S.D.	20%	

a) Portfolio where an investor invests 60% of their wealth in tangent portfolio and rest in risk-free asset.

b) Portfolio where an investor borrows 20% of their wealth at the risk-free rate and invests their wealth and the borrowed funds in the tangent portfolio.

Q1. Calculate expected return and standard deviation

Year	Tangent Portfolio	Risk-Free Asset
E(r)	14%	6%
S.D.	20%	

a) Portfolio where an investor invests 60% of their wealth in tangent portfolio and rest in risk-free asset.

$$E[r_p] = 60\% \times 14\% + 40\% \times 6\% = 10.8\%$$

$$\sigma_p = 60\% \times 20\% = 12\%$$

b) Portfolio where an investor borrows 20% of their wealth at the risk-free rate and invests their wealth and the borrowed funds in the tangent portfolio.

$$E[r_p] = 120\% \times 14\% - 20\% \times 6\% = 15.6\%$$

$$\sigma_p = 120\% \times 20\% = 24\%$$

Q2. The current risk-free rate is 3%, the expected return on the market portfolio is 11%, and Snake security has a beta of 1.2. According to CAPM, what is the expected return on Snake security?

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$$E[r_i] = r_{rf} + \beta_i(r_M - r_{rf})$$

$$\text{Expected return on Snake security} = 3\% + 1.2 (11\% - 3\%) = 12.6\%$$

Q3. Calculate Beta on Paddy’s Portfolio of Stocks

Stock	Price	Number of Shares Held	Beta			
A	\$10	3,000	0.6			
B	\$20	2,000	0.5			
C	\$45	2,000	1.45			
D	\$30	3,000	1.15			
Total						

Q3. Calculate Beta on Paddy’s Portfolio of Stocks

Stock	Price	Number of Shares Held	Beta	Investment	Weight	Beta Contribution
A	\$10	3,000	0.6	\$30,000	0.12	0.072
B	\$20	2,000	0.5	\$40,000	0.16	0.08
C	\$45	2,000	1.45	\$90,000	0.36	0.522
D	\$30	3,000	1.15	\$90,000	0.36	0.414
Total				\$250,000	1.0	<u>1.088</u>

Q4. Leprechaun Asset Management (LAM) will pay a dividend of \$2.00 at the end of year 1. The dividend is expected to grow at 4% in perpetuity. If the risk-free rate is 3%, the expected return on the market portfolio is 11%, and the beta of LAM is 1.3, what is the expected price of this stock?

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$$E[r_i] = r_{rf} + \beta_i(r_M - r_{rf})$$

$$\text{Expected return on LAM stock} = 3\% + 1.3 \times (11\% - 3\%) = 13.4\%$$

$$P = \frac{D_1}{r - g} = \frac{\$2}{13.4\% - 4\%} = \$21.28$$

Q5. Shamrock Company 🍀 is evaluating an expansion of its business. The cash flow forecasts for the project are -\$100 million in year 0 and \$15 million in each of years 1 to 10. The firm's existing assets have a beta of 1.4. The risk free-rate interest rate is 4% and the expected return on the market portfolio is 12%. What is the project's NPV?

Q5. Shamrock Company  is evaluating an expansion of its business. The cash flow forecasts for the project are -\$100 million in year 0 and \$15 million in each of years 1 to 10. The firm's existing assets have a beta of 1.4. The risk free-rate interest rate is 4% and the expected return on the market portfolio is 12%. What is the project's NPV?

$$E[r_i] = r_{rf} + \beta_i(r_M - r_{rf}) = 4\% + 1.4 \times (12\% - 4\%) = 15.2\%$$

$$NPV = -100 + \frac{15}{0.152} \times \left(1 - \frac{1}{(1 + 0.152)^{10}}\right) = -\$25.29 \text{ million}$$

Q6. Calculate Beta on GK Stock

Year	Individual GK Stock Return	Market Portfolio Return
1	15%	8%
2	6%	2%
3	-10%	-7%
4	0%	3%
5	4%	2%

Harvard Management Company Policy Portfolio (1991-2011)

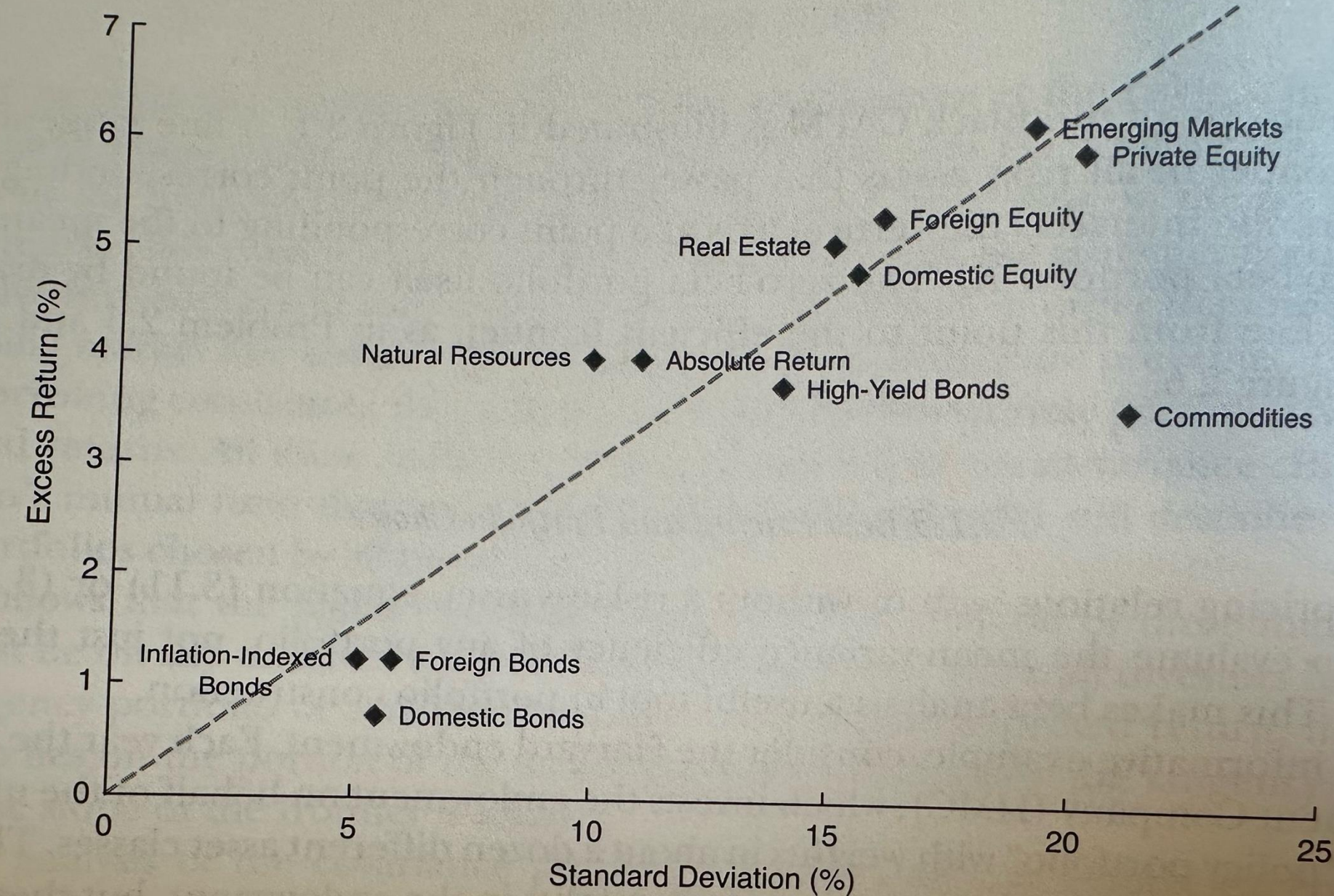


Figure 3.3. Harvard Investment Beliefs: Mean and Standard Deviation

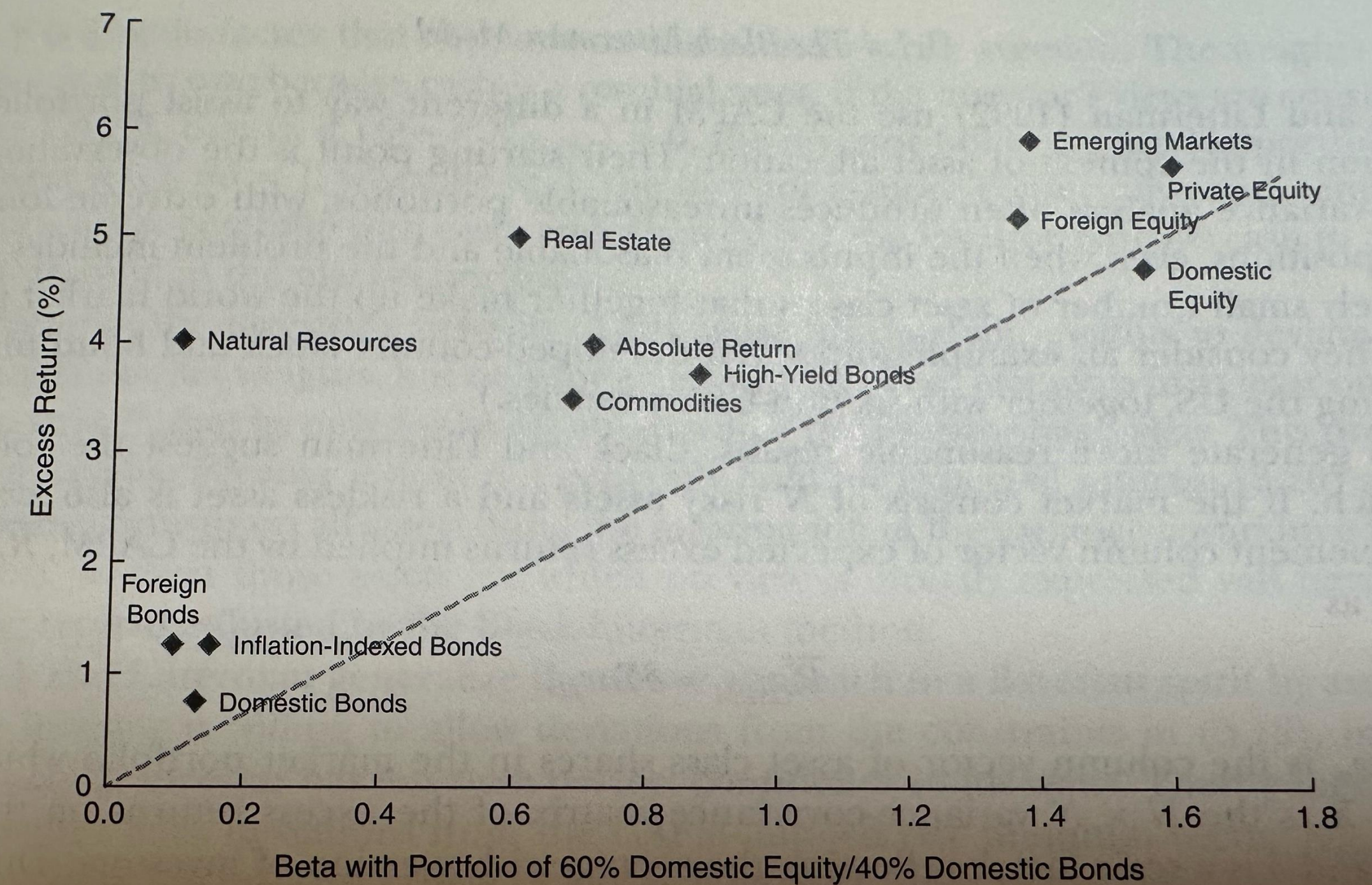


Figure 3.4. Harvard Investment Beliefs: Mean and Beta